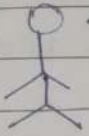
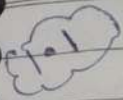


Intro to Machine Learning:

Date _____

Page No. _____



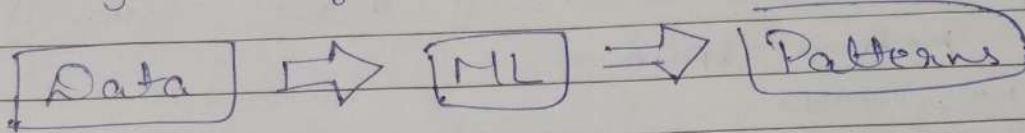
I want to sell car

Problems: Don't wanna put either a high or a low price.

Car	Year	Make	Milage
Car 1	1995	CAZ	200
" 2	1980	VAZ	100
" 3	2016	BWM	5

⇒ Price

Using the age or other value, we'd try to predict



"Features"

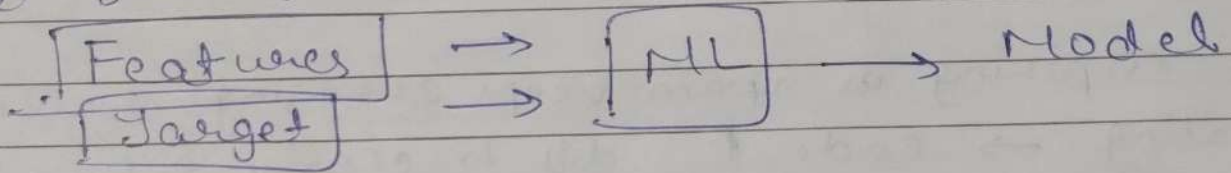
→ what we know

"Target"

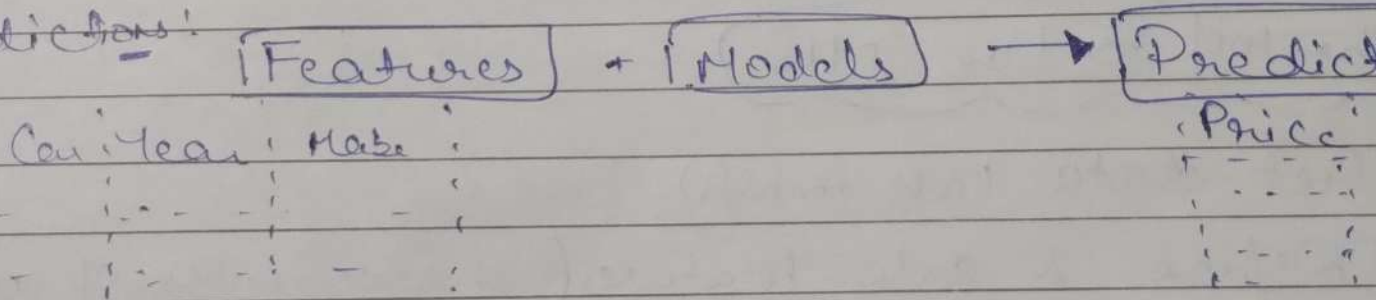
what we want to predict

• ML is process of extracting patterns from data.

Model Training

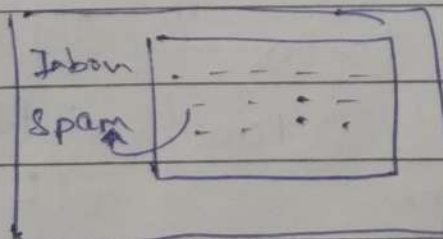


Predictions:

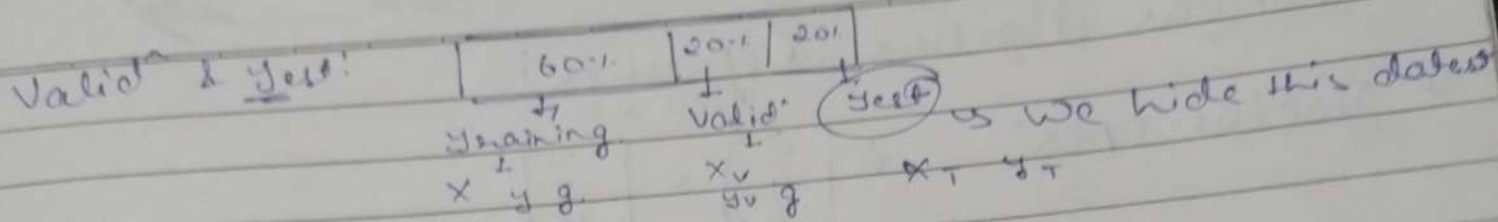


1.2 Machine Learning v/s Rule Based Systems

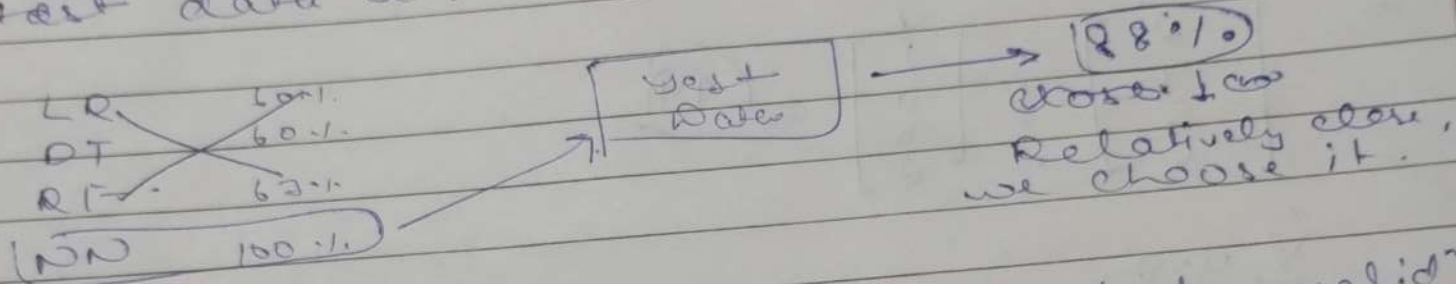
→ Email System:



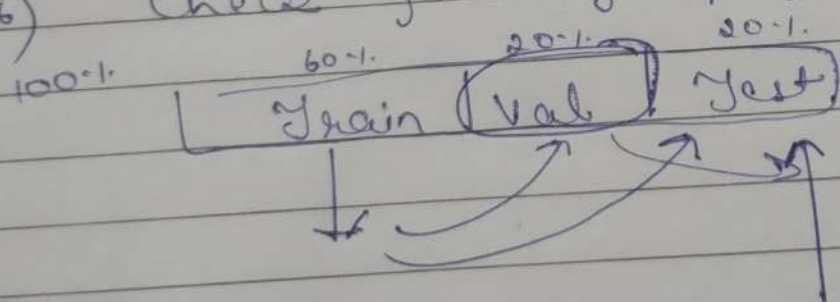
Spam: → Promotions, Discounts, Fraudulent



So ensure, it didn't get lucky we apply NN to test data set.

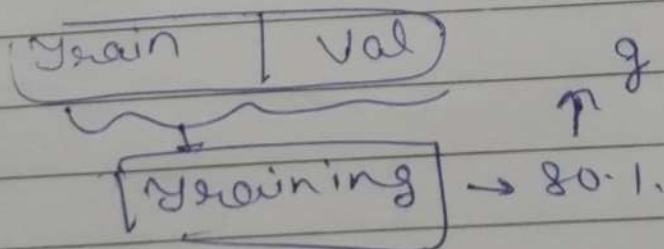


- 1) Split the dataset (in training, test, valid)
 - 2) Train a model
 - 3) Validate using valid
 - 4) Select best model.
 - 5) Apply to test data.
 - 6) Check if everything is good.
- Repeat these steps for diff models.



This dataset is waste (only used for select model)

solⁿ:



	y_v	$\hat{y}_v$	y_v
1)	0.8	1	1
2)	0.7	1	0
3)	0.6	1	1
4)	0.1	0	0
5)	0.9	1	1
6)	0.6	1	0
	Probabilities		Target var (correct in 4/6 cases only)

1 → spam
0 → not spam

> 0.5 = spam

→ correct in $\frac{4}{6}$ cases = 66.6% (Model accuracy)

Req:

g ₁	Logistic Regression	66.6%
g ₂	Decision Trees	60%
g ₃	Random forest	67%
g ₄	NN	80%

→ Best accuracy
(we use this)

* Multiple Comparison Problem: (20% data)

☒ → spam (S) ☐ → Usual (U)

1)	☒	S	S	S	S
2)	☐	S	S	U	U
3)	☒	U	S	U	S
4)	☐	S	S	S	U
		25% cor	50% cor	25%	100%
		LR	RF	DT	NN

(Maybe the model
just got lucky
on this 20% data)

• Have we reached goal?

• Do metrics improve?

Goal: Reduce spam by 50%

- Have we reduced? By how much?
- Evaluate on test group.

→ Do a retrospective: • Was goal achievable?

b) Eval + Deployment: (Often happens together)

- Online eval: deploy the model & evaluate it

→ Iterate: ML projects req many iterations!!

1.5:

Modelling Step: (Model selection process)

July			Aug
x	y	g	$g(\boxed{\text{X}}) = 0.7$

The model from July, didn't see data from Aug, we need to check how this perform in Aug.

$\boxed{80\%}$ $\boxed{20\%}$
Training data set (July)

$\boxed{20\%}$ (Valid data set)
not used in training Aug.

$\boxed{\text{Train}}$
 $\begin{pmatrix} x & y \\ g \end{pmatrix}$

$\boxed{\text{Val}}$
 $\begin{pmatrix} x_{\text{val}} & y_{\text{val}} \\ g(x_{\text{val}}) = \hat{y}_{\text{val}} \end{pmatrix}$

Compare:
 $\hat{y}_v$ & y_v

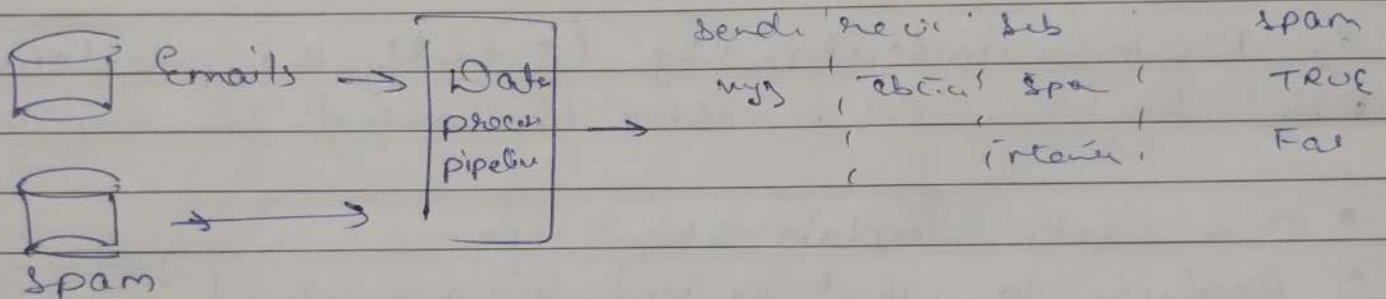


→ Identify the data sources:

- We've a report spam button.
- Is it reliable?
- Is dataset large enough?
- Do we need to get more data?

3) Data Preparation:

- Clean the data.
- Build the pipelines (Raw Data → Clean data)
- Convert into tabular form.



4) Modelling:

- Try diff models & select best one.

Q. Which model to choose?

- Logistic Regression
- Neural networks etc.
- Decision Tree

Sometimes, we may need to go back to data prep:
 + • add new features & fix data issues.

5) Evaluation:

- Measures how well model solves business prob.

1.3:

Supervised Machine Learning:

Features: matrix (X)	Target	Output (Pred)
1) [1, 0, 0, 1]	0	0.93
2) [0, 1, 0, 1]	0	0.25
3) [1, 1, 0, 1]	0	0.5
4) [1, 1, 0, 0]	0	0.15
5) [0, 0, 1, 0]	1	0.97

spam.com $\Rightarrow$ "spam" in 100% cases.

X $\rightarrow$ 2D array $\begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \end{bmatrix}$

y (vector with target val) $\rightarrow$ [1, 0, ...] 1D array

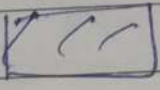
$g(x) \approx y$
Model that takes in matrix X & predicts y

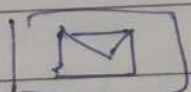
$g(x) \approx y$
Model Feature Target

Types of Supervised ML:

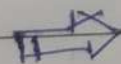
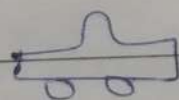
Regression $\rightarrow$ Car $\Rightarrow$ \$50k $g \rightarrow [0, \infty)$

House $\Rightarrow$ \$1M

Classification $\rightarrow$  $\Rightarrow$ Car

 $\Rightarrow$ Spam

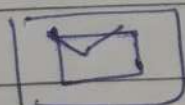
Multi class



Cat
Dog

Car, SUV, Truck

Binary



Spam $\rightarrow$ 1 $g \rightarrow [0, 1]$
Not spam $\rightarrow$ 0

→ Start with rules & use these rules as features

Eg. Sub: Waiting for reply
From: prince@spam.com

We are delighted to inform you've won \$1M.
To claim prize, pay a fee of \$1000.
at paypal.
Congrats!!

↓
Features

- 1) [1, 0, 1, 1] → T & F for features.
2) [0, 1, 0, 1] →
3) [1, 1, 0, 1] →
4) [1, 1, 0, 0] →
5) [0, 0, 1, 0] →

Spam

↓
Target

1

0

0

0

1

Predⁿ (output)

MODEL

Final Dec

Spam

Not "

Not "

Not "

Spam

≥ 0.5 → Spam

Data → Software → Outcome (Hardcoded)
↳ Rule based
Code →

Data → ML → Model
Outcome →
(Spam / Not spam)

Rules:

- If sender = promotions @ spam.com then "spam"
- If title contains "urgent", "lottery" then "spam"
- Otherwise good email

Code: def detect_spam(email):

if email.sender == "promotions @ spam.com":

return spam

if contains(email.title, ('urgent', 'lottery')):

return spam

return good

Problem:

People start complaining abt diff kinds of spam

Rule: → If it contains "deposit money" ⇒ spam

↓
→ Some could use it for valid reasons as well, so solⁿ for

→ Never existing as spam keeps changing, rules keep updating → code ↑, diff to change & manage

Other tool → Use ML

- Get data (all emails)
- Define & calc features (characteristics of mail)
- Train & use model - (classifying spam & non)

eg: Features:

- Length of title > 10 ? T/F
- " " body > 100 ? T/F
- Sender spam.com ? T/F
- Desc contains deposit ? T/F